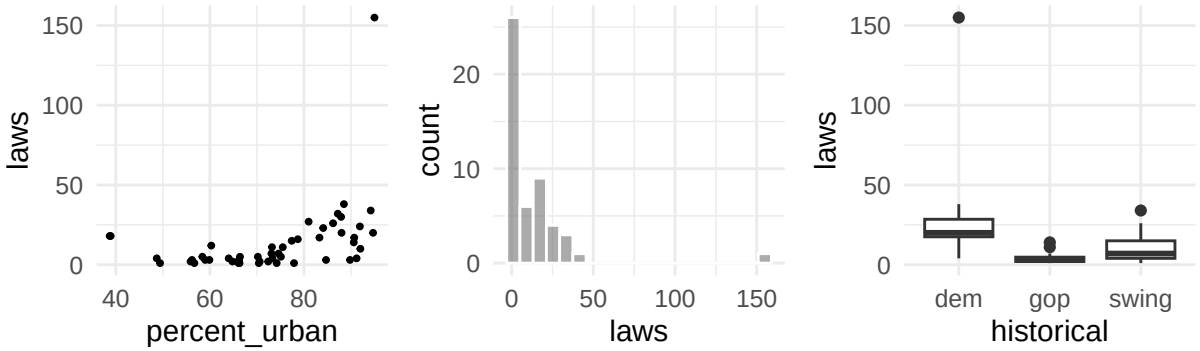
Day 25: Intro to Poisson regression

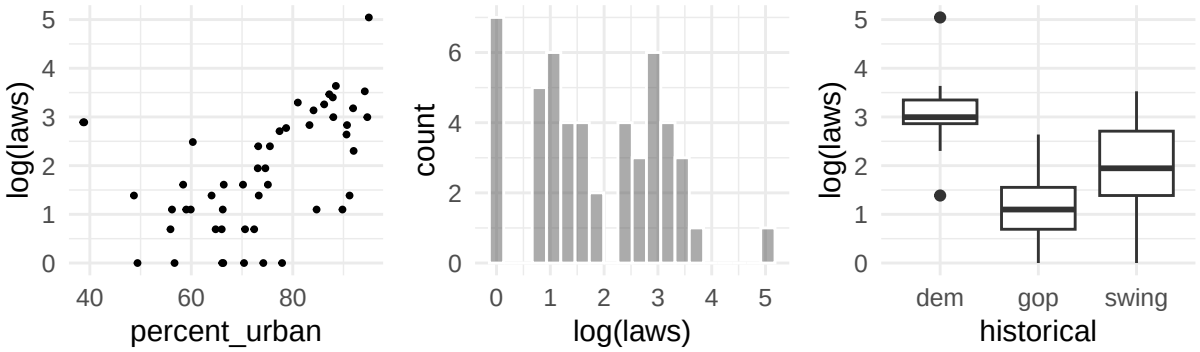
So far, our responses have been quantitative (linear regression) or a success/failure (logistic regression). Today: what if the response is a **count**?

The `equality_index` data (from the `bayesrules` package) records the number of anti-discrimination `laws` in each of the 50 states. We want to model the number of laws using:

- `percent_urban`: the percent of the state's population that lives in urban areas
- `historical`: the state's historical voting pattern in presidential elections: `dem` (typically votes Democratic), `gop` (typically votes Republican), or `swing`

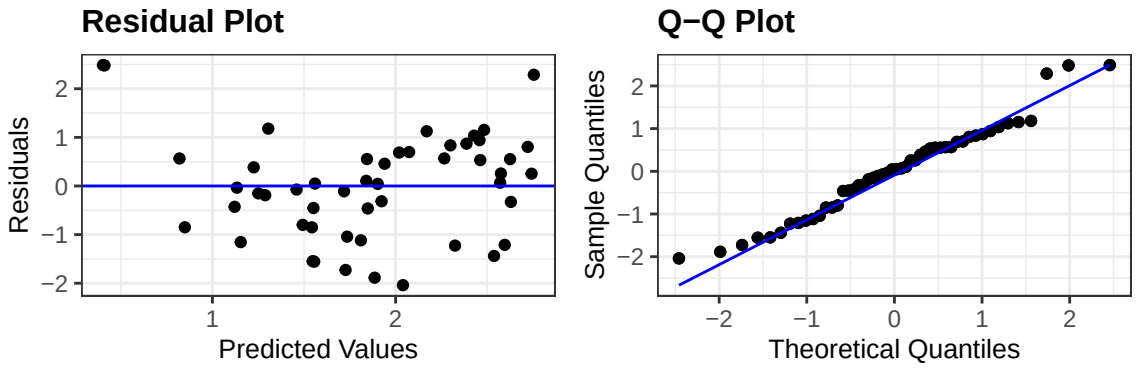


Option 1: transform the response




```
equality_slr <- lm(log(laws) ~ percent_urban, data = equality_index)
```

term	estimate	std. error	t	p-value
(Intercept)	-1.215	0.793	-1.53	0.132
percent_urban	0.042	0.011	3.95	<0.001



Option 2: a model for counts

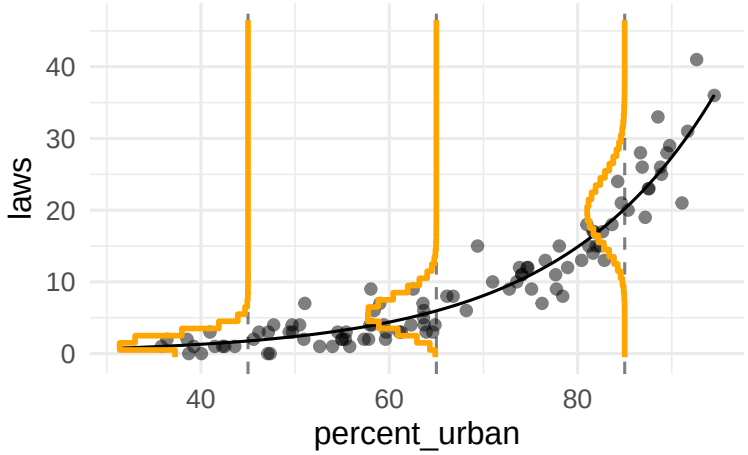
Poisson distribution

The **Poisson distribution** describes the number of events that occur in a fixed interval of time or space, $Y = 0, 1, 2, \ldots$

$$P(Y = y) = \frac{e^{-\lambda}\lambda^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

$E(Y) =$ _____ $Var(Y) =$ _____

The picture below shows simulated data from a Poisson regression model, with the Poisson distribution of Y at three values of x.



Generalized linear models

Linear, logistic, and Poisson regression are all **generalized linear models (GLMs)**, with three parts:

- **Random component:** what distribution does the response follow?
- **Linear predictor:** $\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$
- **Link function:** how do we connect the mean of Y to η?

	Linear regression	Logistic regression	Poisson regression
Response			
Random component			
Mean of Y			
Possible values of the mean			
Link function			

The Poisson regression model

Poisson regression

$$Y_i \sim \text{Poisson}(\lambda_i)$$

Log scale: $\log(\lambda_i) =$ _____ $= \eta_i$
Count scale: $\lambda_i =$ _____

To check whether this model is reasonable, we plot $\log(Y)$ against each quantitative predictor and look for a _____ relationship.

Fitting the model

```
pois1 <- glm(laws ~ percent_urban, data = equality_index, family = "poisson")
```

term	estimate	std. error	z	p-value
(Intercept)	-2.1834	0.3016	-7.24	<0.001

term	estimate	std. error	z	p-value
percent_urban	0.0606	0.0036	16.89	<0.001

Fitted model: $\log(\hat{\lambda}) =$ _____

Interpreting the coefficients

Compare the fitted mean at x and at $x + 1$:

$$\hat{\lambda}(x) = e^{\widehat{\beta}_0 + \widehat{\beta}_1 x} \qquad \hat{\lambda}(x+1) = e^{\widehat{\beta}_0 + \widehat{\beta}_1 (x+1)} = _____$$

Interpreting a slope in Poisson regression

A 1-unit increase in x is associated with the mean count being multiplied by _____. A C-unit increase multiplies the mean count by _____.

Multiple predictors

```
full <- glm(laws ~ percent_urban + historical, data = equality_index, family = "poisson")
```

term	estimate	std. error	z	p-value
(Intercept)	0.7590	0.3269	2.32	0.020
percent_urban	0.0314	0.0037	8.42	<0.001
historicalgop	-1.6121	0.1332	-12.10	<0.001
historicalswing	-0.8288	0.1015	-8.17	<0.001

Example: Interpret the coefficient for `percent_urban` in this model. Why is it different from the coefficient in `pois1`?

Example: Interpret the coefficients for `historicalgop` and `historicalswing`. What is the reference level?

Wald tests and confidence intervals

These work the same way as in logistic regression:

- Test $H_0 : \beta_j = 0$ with $z = \dfrac{\hat{\beta}_j - 0}{SE(\hat{\beta}_j)}$, compared to a $N(0, 1)$ distribution
- A confidence interval for β_j is $\hat{\beta}_j \pm z^* SE(\hat{\beta}_j)$. Exponentiate the endpoints to get an interval for the multiplicative change in the mean count.

Example: Calculate a 95% confidence interval for the `percent_urban` coefficient in `full`, and interpret it for a 10-percentage-point increase.

Summary

- Poisson regression models a **count** response, $Y_i \sim$ Poisson(λ_i), where the mean and the variance are both _____.
- The link function is the _____, so $\log(\lambda_i) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$.
- A 1-unit increase in x_j multiplies the mean count by _____, holding the other predictors constant.
- To check linearity, plot _____ against each quantitative predictor.

R function reference

Function	Purpose
`glm(y ~ x, data, family = "poisson")`	Fit a Poisson regression model
`exp(coef(model))`	Multiplicative change in the mean count for a 1-unit increase in each predictor
`predict(model, newdata, type = "response")`	Predicted mean count ($\hat{\lambda}$)
`confint.default(model)`	Wald confidence intervals for the coefficients (on the log scale)